

Day 4 Hands on subnetting

1. Given a network address of 192.168.10.0/24, divide it into 4 equal subnets and write down the subnet mask, network address, and broadcast address for each subnet.

To divide **192.168.10.0/24** into **4 equal subnets**:

Step 1: Borrow bits

We need **4 subnets**.

Formula:

$2^n \geq \text{Number of subnets}$

$2^2 = 4$, so we borrow **2 bits** from the host portion.

- Original: /24
- Borrowed: **2 bits**
- New prefix: **/26**
- Subnet mask: **255.255.255.192**

Step 2: Find the block size

Last octet of mask = **192**

Block size = $256 - 192 = 64$

So the networks increase by **64**.

Subnet	Network Address	Usable Host Range	Broadcast Address
1	192.168.10.0/26	192.168.10.1 – 192.168.10.62	192.168.10.63
2	192.168.10.64/26	192.168.10.65 – 192.168.10.126	192.168.10.127
3	192.168.10.128/26	192.168.10.129 – 192.168.10.190	192.168.10.191
4	192.168.10.192/26	192.168.10.193 – 192.168.10.254	192.168.10.255

2. For one of the subnets you created, assign valid IP addresses to 5 devices (such as a router, two laptops, a printer, and a mobile) and list which IP goes to which device.

Network: 192.168.10.0/26

- **Subnet Mask:** 255.255.255.192

- **Usable range:** 192.168.10.1 – 192.168.10.62
- **Broadcast:** 192.168.10.63

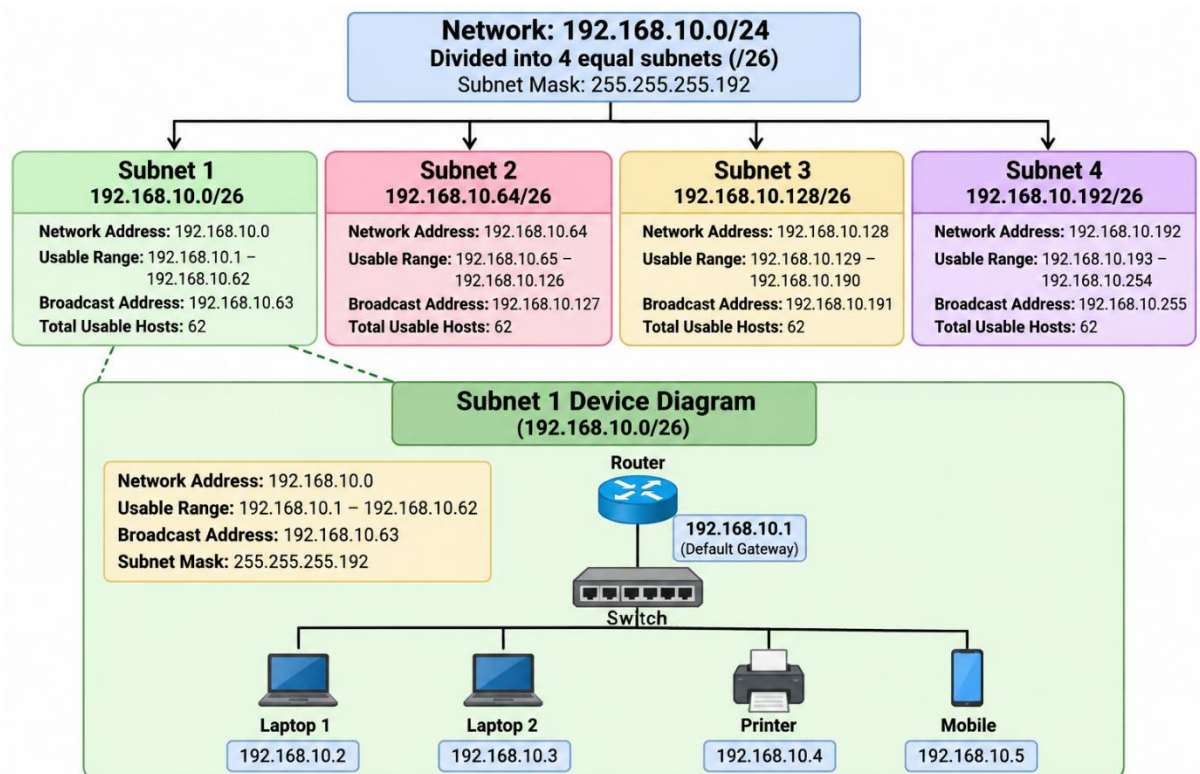
IP Assignment for 5 Devices

Device	IP Address	Subnet Mask
Router	192.168.10.1	255.255.255.192
Laptop 1	192.168.10.2	255.255.255.192
Laptop 2	192.168.10.3	255.255.255.192
Printer	192.168.10.4	255.255.255.192
Mobile	192.168.10.5	255.255.255.192

Default Gateway: 192.168.10.1 for the laptops, printer, and mobile.

All 5 addresses are valid because they are between .1 and .62.

3. Draw a simple diagram (hand-drawn or digital) showing your subnet plan: label each subnet, the range of IPs, and where each device is connected.



4. Document step-by-step how you calculated the subnet mask, number of hosts per subnet, and assigned IP addresses for your plan, as if you are explaining it to a friend who is new to subnetting.

Step-by-Step Subnetting Calculation

We started with the network address **192.168.10.0/24** and needed to divide it into **4 equal subnets**.

Step 1: Calculate the number of bits to borrow

To create 4 subnets, we use the formula:

$$2^n \geq \text{Number of subnets}$$

Here:

$$2^2 = 4$$

Therefore, we need to borrow **2 bits** from the host portion.

The original prefix was **/24**, so:

$$/24 + 2 = /26$$

Therefore, the new subnet mask is:

$$/26 = 255.255.255.192$$

Step 2: Calculate the number of hosts per subnet

A **/26** network leaves **6 host bits** because:

$$32 - 26 = 6$$

The number of total IP addresses is:

$$2^6 = 64 \text{ addresses}$$

Two addresses cannot be assigned to devices:

- One is the **network address**
- One is the **broadcast address**

Therefore:

$$64 - 2 = 62 \text{ usable host addresses per subnet}$$

Step 3: Find the subnet increment

The last octet of the subnet mask is **192**.

We calculate:

$$256 - 192 = 64$$

Therefore, each subnet increases by **64**.

The four subnets are:

Subnet	Network Address	Usable IP Range	Broadcast Address
1	192.168.10.0	192.168.10.1 – 192.168.10.62	192.168.10.63
2	192.168.10.64	192.168.10.65 – 192.168.10.126	192.168.10.127
3	192.168.10.128	192.168.10.129 – 192.168.10.190	192.168.10.191
4	192.168.10.192	192.168.10.193 – 192.168.10.254	192.168.10.255

Step 4: Assign IP addresses to devices

For the device example, we selected **Subnet 1**:

Network: 192.168.10.0/26

Usable range: 192.168.10.1 – 192.168.10.62

Broadcast: 192.168.10.63

We then assigned the first five available IP addresses:

Device	Assigned IP Address
Router	192.168.10.1
Laptop 1	192.168.10.2
Laptop 2	192.168.10.3
Printer	192.168.10.4
Mobile	192.168.10.5

The router's address **192.168.10.1** is used as the **default gateway** for the other devices.

Step 5: Check the addresses

The assigned addresses are all between **192.168.10.1** and **192.168.10.62**, so they are valid host addresses.

We did not assign:

- **192.168.10.0** → Network address
- **192.168.10.63** → Broadcast address

Therefore, the subnet plan is valid and provides **4 equal subnets with 62 usable host addresses in each subnet**.